BB 626: Modeling biological systems and processes

Date : 26 Mar 2025

1. Simulate 10 particles using Brownian dynamics for two cases: (1) No interactions between particles. Only thermal forces. (2) Lennard-Jones (LJ) interactions between particles with $\epsilon = 0.1, 1.0$.

2. Compute and analyze the radius of gyration for all the cases.

   The radius of gyration R_g is given by:

$$R_g = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\mathbf{r}_i - \mathbf{r}_{\mathrm{cm}})^2}, \quad \mathbf{r}_{\mathrm{cm}} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{r}_i.$$

3. Compute and analyze the mean squared displacement (MSD) for all the cases The mean squared displacement (MSD) is:

$$\mathrm{MSD}(t) = \frac{1}{N}\sum_{i=1}^{N}\left\langle(\mathbf{r}_i(t) - \mathbf{r}_i(0))^2\right\rangle.$$